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PROMPT ENGINEERING TASK REPORT

Task 1: Progressive Prompting – Calculator Design

Prompt
Design a simple calculator function. Gradually refine the prompt to include arithmetic operations and error handling. **Explanation**

As the prompt becomes more specific, the calculator logic improves by handling operators correctly and preventing division by zero.

Code
```
def calculator(a, b, op):     if op ==
'+':         return a + b     elif op == '-
':         return a - b     elif op == '*':
return a * b     elif op == '/':         if
b == 0:             return "Error: Division
by zero"         return a / b
    else:
        return "Invalid operator"
```

Output
calculator(10, 5, '+') → 15 calculator(10, 0, '/')

→ Error: Division by zero

```
calculator
15
Error: Division by zero
```

Task 2: Refining Prompts – Sorting Student Marks

Prompt
Sort student marks in ascending order without using built-in sorting functions.

Explanation
Clear prompt instructions lead to a correct comparison-based sorting algorithm that handles duplicate values.

Code
```
def sort_marks(marks):     n =
len(marks)     for i in range(n):
for j in range(i + 1, n):
if marks[i] > marks[j]:
                marks[i], marks[j] = marks[j], marks[i]
return marks
```

Output
Input: [78, 45, 89, 45, 60]

Output: [45, 45, 60, 78, 89]

Task 3: Few-Shot Prompting – Prime Number Validation

Prompt
Check whether a number is prime using few-shot examples.

Explanation
Few-shot examples improve correctness and ensure edge cases like 0 and 1 are handled.

Code
```
def
is_prime(n):
if n <= 1:
        return False     for i in
range(2, int(n ** 0.5) + 1):
        if n % i == 0:
return False     return True
```

Output is_prime(2)

→ True is_prime(4)

→ False is_prime(1)

→ False

```
is prime or not
True
False
```

Task 4: Prompt-Guided UI – Student Grading System

Prompt

Create a user interface that accepts marks and calculates total, percentage, and grade.

Explanation

Prompt clarity results in a structured grading system with accurate calculations.

Code

```
def
grading_system():
    m1 = int(input("Enter marks 1: "))
m2 = int(input("Enter marks 2: "))     m3
= int(input("Enter marks 3: "))
     total = m1 + m2 + m3
percentage = total / 3
     if percentage >= 90:
grade = 'A'     elif
percentage >= 75:
        grade = 'B'
elif percentage >= 60:
        grade = 'C'
else:
        grade = 'Fail'

    print("Total:", total)
print("Percentage:", percentage)
print("Grade:", grade)
```

Output

Input: 80, 75, 85

Output: Total = 240, Percentage = 80, Grade = B

```
grading system
240
80.0
B
```

Task 5: Prompt Specificity – Unit Conversion

Prompt
Convert kilometers to miles and miles to kilometers using clear instructions.

Explanation
Specific prompts improve accuracy and correctness of unit conversion logic.

Code
```
def convert_units(value,
unit):     if unit ==
'km_to_miles':         return
value * 0.621371     elif unit
== 'miles_to_km':
return value / 0.621371
else:
        return "Invalid conversion type"
```

Output
convert_units(10, 'km_to_miles') → 6.21

```
convert units
6.21371
9.977935886933894
```